

Network Monitoring *with* Open Source Tools

Network monitoring is critical to keep computer networks and applications safe from the various cyber attacks that are so common nowadays. This article presents the rationale for network monitoring, along with a few select open source network monitoring tools.



Network monitoring is used to keep an eagle's eye on different computer networks, looking out for slow and failing components. It is quite important, especially in case of sudden unexpected outages and other trouble. Network monitoring is considered to be a part of network management. Enterprise networks have to deal with large data sets being accessed by various devices. Such networks have to provide a reliable and fast service that does not add too much cost to the business.

Networks, nowadays, are far more complex than those of previous years. They traditionally act as the veins of an enterprise by delivering services and data across the organisation. Several technological advances like cloud computing, mobile devices and, more recently, the Internet of Things (IoT) have completely changed the nature of enterprise networks. The demands on today's networks are reliability, speed and uptime – all of which have become more important than ever. Currently, it is more difficult to continuously monitor network infrastructure due to its expansion to cover mobile devices and applications running in the third party cloud environment.

Network monitoring involves different methods that maintain the integrity and security of an internal network or local area network (LAN). Monitoring encompasses software, hardware, spyware, viruses, vulnerabilities like security holes and backdoors, as well as several other aspects that can compromise the integrity of a network.

Basic goals of network monitoring

There are three basic goals for network monitoring, which cover three functional areas (out of a total of five) for network management, based on the Open Systems Interconnection (OSI) model. The remaining two functional areas are configuration and security management which are not related to network monitoring. Let's go through the goals related to network monitoring.

1. **Performance monitoring:** This deals with measuring the performance of a network. There are multiple measurable parameters in a network. But from the list of parameters, relevant ones should be selected from the cost and performance perspective. Such parameters to be measured are referred to as network indicators since they indicate the performance attributes of the network. Some of the